MIL-P-48780(PA)
27 January 1975
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

PRACTICE WARHEAD SECTION: M65E4

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of warhead section entitled Practice Warhead Section: M65E4 which is used with the Sergeant warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICAITONS

MILITARY

MIL-S-48802(PA) - Simulator Assembly Warhead Section: M3E3.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460 - Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

FSC: 1115

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DRAWINGS

ORDNANCE CORPS

8829875	- Skin.
8829881	- Joint Ring Assembly.
8830019	- Compartment Assembly Case Section, Ballistic Rear.
8838463	- Container 9125576 (Modified).
8838468	- Fire Plug Assembly.
8838469	- Safe Plug Assembly.
8838621	- Forward Ballast.
8838629	- Fuze Skin Assembly.
8838635	- Retaining Block.
8838754	- Static Tube, Training: M10.
8857123	- Probe Storage Container.
8857145	- Cable Assembly W15L.
8857146	- Check Plug Assembly.
8857683	- Practice Warhead Section: M65E4
8857848	- Ballast Rear.
8857916	- Connector and Shell Assembly.
8857917	- Simulator Assembly, Warhead Section: M3E3.
8858044	- Electrical Schematic M3E3 Simulator, Warhead Section.
9125032	- Aft Ballast Assembly.
9125576	- Fare Section Container M421.
10013351	- Structure Assembly Compartment Warhead.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The Practice Warhead M65E4 shall comply with all requirements specified on the drawing (dwg) 8857683, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

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3.4 Serialization. A six letter serial number shall be assigned to the item. Serial numbers required on these items shall be obtained by means of a letter addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial numbers required.

3.5 Physical characteristics. Unless otherwise specified all tests shall be performed at a temperature of plus (+) 80 degrees (°) plus or minus (±) 20 degrees Fahrenheit (°F).

3.5.1 Weight. The complete Practice Warhead Section M65E4 with Static Tube: M10 attached shall weigh 1611 pounds (lbs) ± 20 lbs when determined as specified in 4.4.1.

3.5.2 Center of gravity. The center of gravity of the complete Practice Warhead Section M65E4 with Static Tube M10 attached shall fall between Station 97.0 and 111.0 when determined as specified in 4.4.2.

3.6 Operational characteristics. Unless otherwise specified, all tests shall be conducted at 80 ± 20°F.

3.6.1 Insulation resistance. The insulation resistance between the pins of connectors as designated in Table I shall be 20 megohms minimum when tested as specified in 4.4.3. The Safe Plug 8838469 shall be installed for this test.

Table I

Connector P919

From Pin	To Pins
4	9, 13, 15, 24, 35, 36, 50, 62, Case
9	4, 13, 15, 24, 35, 36, 50, 62, Case
13	4, 9, 15, 24, 35, 36, 50, 62, Case
15	4, 9, 13, 24, 35, 36, 50, 62, Case
24	4, 9, 13, 15, 35, 36, 50, 62, Case
35	4, 9, 13, 15, 24, 36, 50, 62, Case
36	4, 9, 13, 15, 24, 35, 50, 62, Case
50	4, 9, 13, 15, 24, 35, 36, 62, Case
62	4, 9, 13, 15, 24, 35, 36, 50, Case
Case	4, 9, 13, 15, 24, 35, 36, 50, 62

Connector J16

From Pin A	To Case

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3.6.2 Circuit resistance and continuity. The resistance between connector pins shall be as designated when the stepping switch is in the position indicated in Table II. Pulse the stepping switch with 28 direct current voltage (vdc) plus or minus (±) 10 percent (%) with a pulse width of less than one second. The positive pulse shall be applied to P919-31 and the return shall be connected to P919-18. Test as specified in 4.4.4. The Fire Plug 8838468 shall be inserted for this test.

Table II

Ledex Position	Pin to Pin		Resistance (ohm)	Tolerance (±%)
	P919-9	P919-36	0.2 max.	
	P919-9	P919-42	0.2 max.	
	P919-9	P919-3	0.2 max.	
	P919-21	P919-14	0.2 max.	
	P919-21	P919-27	0.2 max.	
	P919-21	P919-40	0.2 max.	
	P919-21	P919-1	0.2 max.	
	P919-21	P919-8	0.2 max.	
	P919-15	P919-22	715	10
	P919-18	P919-24	10 min.	
	P919-11	P919-4	290	10
	P919-11	P919-17	440	10
	P919-24	P919-31	0.2 max.	
	P919-50	P919-70	0.2 max.	
	J16-A	J16-B	0.2 max.	
	J16-A	J16-C	0.2 max.	
1	P919-28	P919-35	30,000	10
1	P919-62	P919-68	11,000	0.1
1*	P919-50	P919-69	3,000	0.1
2*	P919-50	P919-68	3,000	0.1
3*	P919-50	P919-68	5,000	0.1
4*	P919-50	P919-68	7,000	0.1
5*	P919-50	P919-68	9,000	0.1
6*	P919-50	P919-68	9,000	0.1

*Apply 28 vdc to pins P919-31 and P919-18 to pulse Ledex.

3.6.3 Operation test. Each warhead section shall be operated and shall function as specified when tested as specified in 4.4.5. Switch position indication shall be as specified in Table II.

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3.7 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.8 Simulator M3E3. The simulator assembly shall comply with requirements specified in dwg 8857917 and MIL-S- (PA).

3.9 Workmanship.

3.9.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.9.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.9.3 Burrs. Parts shall be free from burrs.

3.9.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.9.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.9.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 1 complete assembly (see dwg 8857683) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table III. Tests shall be conducted in order specified in Table III.

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Table III

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Weight (see 3.5.1)		X
3. Center of Gravity (see 3.5.2)		X
4. Insulation resistance (see 3.6.1)		X
5. Circuit resistance (see 3.6.2)		X
6. Operation (see 3.6.3)		X
7. Simulator (see 3.8)		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table IV and Table V), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table IV
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table V
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Rear ballistic case section compartment assembly (see dwg 8830019).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Any components missing, loose or damaged	Visual
102.	Evidence of poor workmanship (see 3.9)	Visual
103.	Objective evidence relating to material missing or improper (see 3.1)	Visual

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Minor:

201.	More than 4.0 percent of rivets and lockbolts not flush with skin within 0.005 inches	Gage
202.	Scratches through protective finish with base metal exposed	Visual

4.3.2.1.2 Forward ballast (see dwg 8838621).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Finished outside (small) diameter incorrect	Gage
102.	Mislocation of handling equipment mounting holes	Gage
103.	Handling equipment mounting holes oversize	Gage
104.	Evidence of poor workmanship (see 3.9)	Visual
105.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
201.	Scratches in protective finish with base metal exposed	Visual

4.3.2.1.3 Fuze skin assembly (see dwg 8838629).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Contour of skin functionally incorrect	Gage
102.	Mating diameter at Station 54.00 incorrect	Gage
103.	Bolt hole diameter oversize	Gage
104.	Retaining block will not accept static tube	Gage
105.	Any components missing, loose or damaged	Visual

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106. Evidence of poor workmanship (see 3.9).....Visual
107. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor:
201. Scratches through protective coating with base metal exposed.....Visual

4.3.2.1.4 Static tube, dummy, M10 (see dwg 8838754).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Will not mount in retaining block	Gage
	102. Any components missing, loose or damaged	Visual
	103. Evidence of poor workmanship (see 3.9)	Visual
	104. Marking "TOP" on wrong fin	Visual
	105. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Scratches in protective finish with the base metal exposed	Visual
	202. Incorrect or improper marking	Visual

4.3.2.1.5 M65E4 Practice warhead section (see dwg 8857683).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Eccentricity exceeds 0.200 inches T.I.R. from Station 8.125 to Station 139 (Installation of ballast optional)	Gage
	102. Mismatch at Station 8.125, 54.75 and 55.0 exceeds 0.020	Gage
	103. Gap in skin contour at Station 8.125, 54.75, and 55.0 exceeds 0.020	Gage

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104. Overall length tolerance of Assembly from Station 8.125 to 139.0 exceeds ± 0.200 inches.....Gage
105. Incorrect mating of connectors.....Visual
106. Any component missing, loose or damaged.....Visual
107. Protruding fasteners.....Visual
108. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor:
201. Bare spot in protective coating except for slight scratches.....Visual
202. Improper or illegible markings.....Visual

4.3.2.1.6 Ballast rear (see dwg 8857848).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Location of mounting holes	Gage
	102. Diameter, mounting holes	Gage
	103. Location of handling equipment holes	Gage
	104. Diameter, handling equipment holes	Gage
	105. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Bare spot in protective coating except for slight scratches	Visual
	202. Evidence of poor workmanship (see 3.9)	Visual

4.3.2.1.7 Simulator, warhead section: M3E3 (see dwg 8857917).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Mounting bolt holes not functionally located	Gage

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102. Any components missing, loose or damaged.....Visual
103. Evidence of poor workmanship (see 3.9).....Visual
104. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor:
201. Scratches in protective finish with base metal exposed.....Visual

4.3.2.1.8 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Preservation, packing, and packaging not in accordance with (see dwg 8838463)	Visual
	102. Marking not in accordance with MIL-STD-1169	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.9)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

	Test	Test Classification
a.	Weight	Major
b.	Center of gravity	Major
c.	Insulation resistance	Major
d.	Circuit resistance	Major
e.	Operation	Critical

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Weight, center of gravity, insulation resistance, circuit resistance and operation. Each Practice Warhead Section in the lot shall be tested in accordance with 4.4.1 through 4.4.5. If any item fails to comply with 3.5.1 through 3.6.3, the item shall be rejected and removed from the lot.

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4.3.4 Acceptance inspection equipment.

4.3.4.1 Major and Minor characteristics and tests. Inspection and testing shall be performed with the acceptance test equipment, operating instructions and calibration instructions prepared by the contractor. The contractor shall obtain approval of such designs, operating instructions and special calibration instructions as specified in 6.3 prior to use on the contract.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Weight. The total weight of the M65E4 Practice Warhead Section shall be determined by weighing on a suitable scale.

4.4.2 Center of gravity. The center of gravity of the M65E4 Practice Warhead Section shall be determined by a suitable means using suitable gages and measuring devices.

4.4.3 Insulation resistance. Insulation resistance shall be measured by means of an insulation resistance tester at 500 ± 50 vdc. The maximum current output of the insulation resistance tester shall not exceed 5 milliamperes. All selector switches shall be in Position 1 for this test unless otherwise specified. (See Table I and II.)

4.4.4 Circuit resistance. A suitable electrical resistance measuring instrument shall be connected to the specified connector pins for the measurement of circuit resistance. (See Table II).

4.4.5 Operation test. Step the stepping switch through least one complete revolution. The switch shall be stepped and monitored as indicated in Table II.

Connect a 680 ohm ± 10% resistor to pins P919-8 and P919-11. Connect a Bendix Barosetter type 8821748 to the appropriate terminals on P919 as shown in Figure 1. Set the Barosetter at -1300 ft. Press the setting button until a null in indicated (a null is indicated on the Bendix Barosetter when the green "null" light comes on). Repeat the above for settings of 10,000 ft. and 25,000 ft. Measure the voltage between pins P919-8(+) & P919-11(-). This voltage shall measure 46 ± 6 vdc as long as the Barosetter is connected and power is applied.

> CAUTION: 680 ohm resistor must be connected to pins P919-8 and P919-11 before any power is applied.

Disconnect Barosetter. Connect the positive side 150 vdc ± 10% to pin P919-8. Connect a 680 ohm ± 10% resistor from

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pin J14-A to the negative side of the 150 vdc. Measure the voltage across the 680 ohm resistor. The measured voltage across the 680 ohm resistor shall not exceed 5.0 millivolts.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. P;servation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The Practice Warhead Section: M65E4 shall be packaged in accordance with dwg 9125576 as modified in accordance with dwg 8838463 using the specified moisture barrier.

5.1.2 Level C. The Practice Warhead Section: M65E4 shall be packaged in accordance with dwg 9125576 as modified in accordance with dwg 8838463 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The Practice Warhead Section: M65E4 shall be packed in accordance with dwg 9125576 as modified in accordance with dwg 8838463.

5.3 Marking. Marking shall be as specified in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Supersession data. This specification supersedes the requirements of Purchase Description X-PA-PD-2265 dated 19 December 1962.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1115-A053

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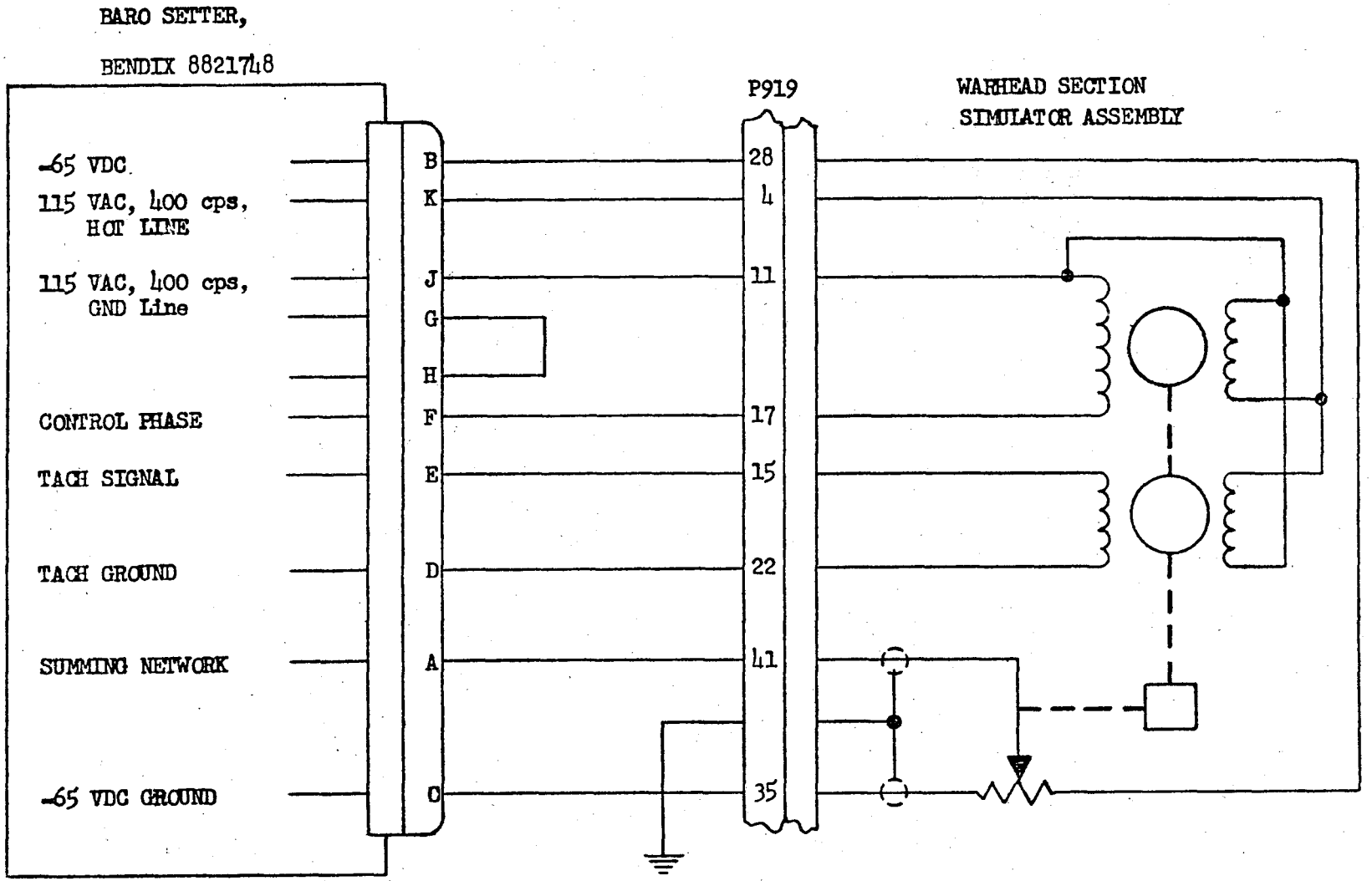


Figure 1. Test Circuit

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